MIL-C-38351A(USAF)
22 August 1972
SUPERSEDING
MIL-C-38351(USAF)
18 March 1965

MILITARY SPECIFICATION

CLOTH, PARACHUTE, NYLON, AROMATIC, NONMELTING

1. SCOPE

1.1 Scope. This specification covers requirements for two types and three classes of aromatic polyamide cloth.

1.2 Classification. The aromatic polyamide cloth shall be of the following types and classes, as specified (see 6.2).

Type I, class 1	-	4.7 oz.
Type I, class 2	-	6.5 oz.
Type I, class 3	-	12.0 oz.
Type II, class 1	-	7.0 oz.
Type II, class 2	-	18.0 oz., resin treated

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-P-1133	Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

Federal

Fed Std No. 4	Glossary of Fabric Imperfections
Fed Std No. 191	Textile Test Methods

FSC 8305

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Military
MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Federal Trade Commission

Textile Fiber Products Identification Act, effective March 3, 1960.

(Copies may be obtained without charge from the Federal Trade Commission, Washington D. C. 20580.)

3. REQUIREMENTS

3.1 Yarn. The yarn shall be a 200 ± 15 denier, nonmelting, aromatic polyamide. The yarn shall not char at a temperature below 675°F when tested as specified in table II.

3.2 Construction and physical properties. The construction and physical properties of the finished cloth shall be as specified in table I and 3.3 through 3.10.

3.3 Width. The width, inclusive of selvage, for all types and classes of cloth shall be as specified by the procuring activity (see 6.2).

3.4 Weave

3.4.1 Type I, class 1. The weave shall be 2-up, 2-down, right-hand twill.

3.4.2 Type I, classes 2 and 3. The weave shall be as shown in figure 1.

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Table I. Construction and physical properties

Requirement	Type I			Type II	
	Class 1	Class 2	Class 3	Class 1	Class 2
Weight (oz/yd^2) max	4.7	6.7	12.0	7 0	18.0
Yarns/inch, min					
Warp	74	63	38	60	59 1/
Filling	74	48	38	45	60 1/
Ply, min					
Warp	Singles	2	5	2	5
Filling	Singles	2	5	2	5
Twist (turns/inch) min					
Warp	4.0	2.5	2.5	2.5	2.5
Filling	4.0	2.5	2.5	2.5	2.5
Breaking strength (lbs/inch) min					
Unaged					
Warp	165	265	425	250	950 1/
Filling	165	200	425	190	950 1/
After Heat Aging (% of unaged)					
Warp	85	85	85	85	85 1/
Filling	85	85	85	85	85 1/
Ultimate elongation, percent min					
Warp	30	25	15	35	-
Filling	30	25	15	25	-
Tearing strength, lbs, min					
Warp	14	20	75	15	-
Filling	14	20	75	15	-
Thickness, inch, max	0.011	0.020	0.032	0.015	0.036
Air permeability ($ft^3/min/ft^2$)	40 to 70	50 to 80	40 to 80	8 max	-

1/ The cloth prior to the resin treatment shall meet the requirements of table I, for type II, class 2.

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3.4.3 Type II, class 1. The weave shall be plain 1-up and 1-down.

3.4.4 Type II, class 2. The weave shall be 3 by 4 basket (3 ends weaving as 1, 4 picks per shed).

3.5 Finish (type II, class 2 only). The type II, class 2 cloth shall have a stiffness of 0.45 to 0.65 inch pounds in the warp and 0.65 to 0.85 inch pounds in the filling when tested as specified in 4.4 (see 6.4).

3.6 Color. Unless otherwise specified (see 6.2), the color of the cloth shall be Sage Green 1565. The color shall be obtained by the use of melt spun solution dyed yarn.

3.7 PH. The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.0 when tested as specified in 4.4.

3.8 Length and put-up

3.8.1 Type I, classes 1 and 2, and type II, class 1. The cloth shall be put up in rolls in accordance with PPP-P-1133. Each roll shall contain not less than 80 yards nor more than 200 yards. Each roll, 100 yards or less, shall contain not more than 2 pieces, and each roll, over 100 yards, shall contain no more than 3 pieces. Each piece shall contain not less than 40 yards.

3.8.2 Type I, class 3, and type II, class 2. The cloth shall be put up in rolls in accordance with PPP-P-1133. Each roll shall contain not less than 80 yards nor more than 120 yards. Each roll shall contain not more than 2 pieces and each piece shall contain not less than 40 yards.

3.9 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act, effective March 3, 1960.

3.10 Face identification. The face side of the cloth shall be identified by stamping on that side the word "Face" at each end of the roll.

3.11 Workmanship. The cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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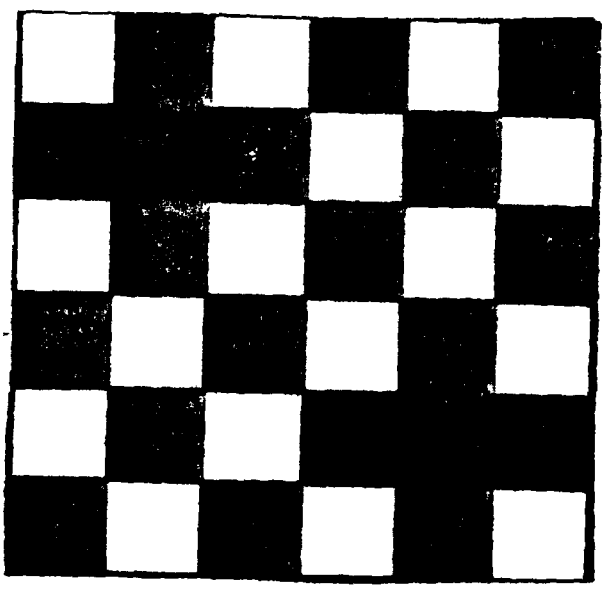


Figure 1. Weave Design (Type I, Classes 2 and 3)

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4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection for acceptance. Inspection shall be in accordance with MIL-STD-105, except where otherwise indicated.

4.2.1 Testing of component. The component listed in table II shall be tested for the characteristics specified in accordance with the referenced test method of Fed Std No. 191 except where otherwise specified. The lot size shall be expressed in pounds, and the sample unit shall be 500 yards of the yarn. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The sample size shall be as follows.

Lot size (pounds)	Sample size
1000 or less	3
1001 to 5000	5
5001 and over	7

Table II. Testing of component

Component	Characteristic	Requirement reference	Test method of Fed Std No. 191 or test reference
Yarn	Denier	3.1	4021
	Ply	Table I	Visual 1/
	Twist	Table I	4054
	Carbonization	3.1	1534 2/

1/ One determination per sample unit reported as pass or fail.
2/ The supplier's certificate of compliance showing that the yarn identified in 6.3 was used may be submitted in lieu of test method.

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4.2.2 Examination of the end item

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage of the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in Section III of Fed Std No. 4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following.

Lot Size (Yards)	Sample Size (Rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls, in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 20.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 30 points. If no individual roll exceeds the point level, the lot shall be acceptable with respect to this characteristic. If one roll exceeds the point level, a second sample equal in size to the first, shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceeds the point level. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 sq yds}$$

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4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to and including 3 inches in any dimension.

For defects exceeding 3 inches, but not exceeding 6 inches in any dimension.

For defects exceeding 6 inches, but not exceeding 9 inches in any dimension.

For defects exceeding 9 inches in any dimension

NOTE The following defects, when present, shall be scored 4 points for each yard in which they occur.

Color other than Sage Green 1565.
Baggy, ridgy, or wavy cloth
Objectionable odor
Width less than specified.

4.2.3 Examination for length

4.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum or more than the maximum specified, or more than 2 yards below the length marked on the ticket, or any roll found to contain more than the specified number of pieces, or any one piece less than 40 yards in length shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual length of the rolls in the sample is less than the total of lengths marked on the tickets.

4.3 Examination of preparation for delivery requirements An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking requirements comply with Section 5 of this specification and that fiber identification is in accordance with 3.9

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Table III. Test methods

Characteristic	Requirement reference	Test method of Fed Std No. 191 or test reference	Number of determinations per indiv sample unit 6/	Results reported as 6/
Yarns per inch 1/	Table I	5050	-	-
Thickness 1/	Table I	5030	-	-
Weight 1/ 5/	Table I	5041	-	-
Breaking strength (Unaged) 2/ 3/	Table I	5104	-	-
Breaking strength (Unaged) 1/ 2/ 4/	Table I	5100	-	-
Breaking strength (Aged) 1/ 2/	Table I	4.4.2 and 5100 or 5104	5	% of unaged
Elongation 3/	Table I	5104	-	-
Tear resistant 3/	Table I	5134	-	-
Air permeability 3/	Table I	5450	-	-
Weave	3 4	Visual	1	Pass or fail
Stiffness 4/	3.5	5202	-	-
pH	3.7	2811	-	-

1/ For type II, class 2 cloth, this test shall be performed on the cloth prior to the resin treatment

2/ Breaking strength test specimens for the unaged and the oven aged tests shall be prepared in a way so that both contain the same warp thread in the case of the warp specimens and the same filling threads in case of the filling specimens.

3/ Applicable to type I, classes 1, 2, and 3, and type II, class 1.

4/ Applicable to type II, class 2 only.

5/ For type II, class 1, the results shall be reported to the nearest 0 01 oz.

6/ When the data in the "number of determinations per sample unit" and the "results reported as" columns are not specified, they shall be as required by the referenced test method.

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4.4 Testing of the end item. The methods of testing specified in Fed Std. No. 191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The lot shall be unacceptable if one or more units fail to meet any requirements specified except for the permeability test (see 4.4.1). The lot size shall be expressed in terms of 1 yard each. The sample size shall be as given under 4.2.1. The sample unit for all type I, and class 1 of type II cloth shall be 2 continuous yards, full width, of the finished cloth. The sample unit for type II, class 2 cloth shall be 1-1/2 continuous yards, full width of the cloth prior to resin treatment, and one-half yard, full width, of the treated cloth.

4.4.1 Air permeability. A 2-yard sample, the full width of cloth, shall be selected from each shipper's roll and shall be tested for air permeability as specified in table III. Failure to meet the air permeability requirement shall cause rejection of the roll of cloth represented

4.4.2 Breaking strength after aging. The size of the specimens shall be the same as specified in the unaged tests (see table III). The specimens shall be placed in an oven at 500° ± 10°F for 24 hours. Upon removal, the specimens shall be conditioned at standard conditions for 4 hours and tested for breaking strength as specified in table III. The breaking strength after the aging treatment shall be reported as percent of the unaged specimens.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Levels A, B, and C. The cloth, put up in rolls, shall be packaged in accordance with the applicable requirements for class 2 fabrics of PPP-P-1133.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

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5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

5.4 Age. The cloth shall be not more than 2 years old from the date of manufacture to the date of delivery.

6. NOTES

6.1 Intended use. The cloth is intended for use in the manufacture of parachute canopies, packs, and pack stiffeners.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of cloth required (see 1.2).
c. Width and quantity required (see 3.3).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Nomex (formerly called HT-1), manufactured by E. I. duPont, is an acceptable yarn.

6.4 Finish. The type II, class 2 cloth has been made successfully by using the following finishing process·

Scour:	180°F for 30 minutes in a jig
	1 g/l Duponal RA
	1 g/l TSPD
	Overflow rinse until clear
Dry:	Minimum tension on tenter frame (250°F)
Finish.	Pad at 80°F, 2 dips; 2 nips to 30 percent wet pickup
	660 g/l Resloom M-75
	10 g/l Diammonium phosphate
Dry·	Minimum tension on tenter frame (250°F)
Cure·	Minimum tension on tenter fram (400°F for 3 minutes)
Wash.	Crab one pass over one roll
	2 g/l sodium perborate at 80°F
	Crab one pass over one roll
	Clear water at 80°F
Dry:	Minimum tension on tenter frame (250°F).

Custodian:
Air Force - 11

Preparing activity:
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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: MIL-C-83351A(USAF)	2. DOCUMENT DATE (YYMMDD): 720822

3. DOCUMENT TITLE: CLOTH, PARACHUTE, NYLON, AROMATIC, NONMELTING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

[redacted]

6. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
SA-ALC/TIRDI	(512) 925-6314	945-6314
c. ADDRESS (Include Zip Code): Kelly AFB TX 78241-5000	IF YOU DID NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office, 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466, Telephone: (703) 756-2340, AUTOVON 289-2340	

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